

application implementation

- They don't carry any executable file that can be installed or removed from Android mobile devices

Hybrid apps: Hybrid applications are neither native apps nor Web apps. They are implemented with Web technologies and packaged as applications for distribution. These apps can access native device features and APIs. Basically, a hybrid app is a native mobile application, which hosts a Web browser control within its main UI screen. Here are the imperative characteristics of an Android mobile hybrid app:

- UI implementation using Web technologies (HTML, CSS and Java Script)
- Apps are capable of using Android mobile platform features
- An executable file installs and resides at the mobile device
- Apps can be distributed via the Android platform application store (Google Play) or an enterprise distribution mechanism

Figure 2 depicts an analogy of a hybrid application. Broadly, it has two layers – the container and the user interface (UI). The container is implemented with Android native mobile technology, whereas the UI is implemented with Web technologies (HTML, CSS and JS).

There are many frameworks and tools for Android mobile hybrid application development. The main selling point for hybrid applications is cross-platform development. The following are the features which can be leveraged with hybrid applications:

- Integration of open source frameworks with HTML5
- Liquid layouts for multi-screen UIs
- Local storage, multimedia handling, semantics and forms, graphics, etc.
- A single code based architecture model for multi-platform presentations
- Hybrid framework that bundles the HTML5 based view layer, with native platform containers, to create deployable builds

Other applications

There are other distinctive ways of information communication in Android mobile devices, like SMS and IVR (interactive voice response).

SMS based apps: This is the short-code based SMS application. In this application, users are required to send SMSs in specified short-code to avail services. These kinds of applications are primarily driven by network operators and the SMS server, to extend the application services for feature phones.

IVR apps: IVR applications are used for automated interactions with callers, typically, in the customer support department to offer services 24x7. These applications are multi-lingual in order to serve customers across

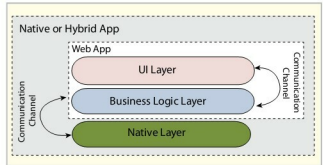


Figure 1: Application layer architecture

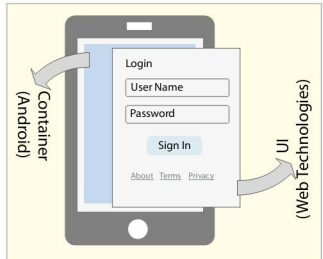


Figure 2: Hybrid app

heterogeneous regions. IVR apps are driven by network operators and IVR servers. They don't require any additional client application to be installed on Android mobile phones.

Application layer structure

Android mobile applications incorporate multiple layers in one application codebase.

Figure 1 illustrates three distinctive layers.

- **UI layer:** This represents the UI/UX and view implementation for Android mobile applications. Android platform has multiple ways of implementing the UI layer as hybrid/Web or native (Android Activity). The hybrid and Web application UI layer is implemented with HTML, CSS, JS and other UI frameworks/tools (jQuery, jQueryMobile, LESS, Bootstrap, etc). This is the layer where platform-specific UI/UX guidelines are required to be implemented as per application features.
- **Business logic layer:** This layer is responsible for core logic and implementation handling for applications. It also communicates with the server for the required data and information. Android-recommended implementation approaches and the respective design patterns can be used to implement application